MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 10** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 1: Real Numbers** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

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Q1. If $\text{LCM}(x, 18) = 36$ and $\text{HCF}(x, 18) = 2$, then the value of x is: [1]

(a) 2 (b) 3 (c) 4 (d) 6

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) 4

We know the product relation:

Formula Used: $x \times 18 = \text{HCF}(x, 18) \times \text{LCM}(x, 18)$

$18x = 2 \times 36 = 72 \implies x = \dfrac{72}{18} = \boxed{4}$. [1M]

Q2. If $\text{HCF}(a, 8) = 4$ and $\text{LCM}(a, 8) = 24$, then a is equal to: [1]

(a) 12 (b) 16 (c) 6 (d) 8

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 12

We know the identity:

Formula Used: $a \times 8 = \text{HCF}(a, 8) \times \text{LCM}(a, 8)$

$8a = 4 \times 24 = 96 \implies a = \dfrac{96}{8} = \boxed{12}$. [1M]

Q3. For what natural number n does 4^n end with the digit zero? [1]

(a) for all even n (b) for $n = 5$
(c) for no natural number n (d) for $n = 10$

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (c) for no natural number n

Prime factorising 4^n:

Formula Used: $4^n = (2^2)^n = 2^{2n}$ (prime factor 5 is absent)

Since ending with 0 requires both 2 and 5 as prime factors, 4^n ends with 0 $\boxed{\text{for no natural number } n}$. [1M]

Q4. If n is any natural number, then 12^n cannot end with the digit: [1]

(a) 2 (b) 4 (c) 6 (d) 0

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (d) 0

Prime factorising 12^n:

Formula Used: $12^n = (2^2 \times 3)^n = 2^{2n} \times 3^n$

Powers of 12 cycle through unit digits $2, 4, 8, 6$ ($12^1 = 12, 12^2 = 144, 12^3 = 1728, 12^4 = 20736$), and prime 5 never divides 12^n, so it can NEVER end with $\boxed{0}$. [1M]

Q5. If the LCM of two natural numbers is 3600, which of the following numbers CANNOT be their HCF? [1]

(a) 600 (b) 500 (c) 400 (d) 150

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) 500

We know the divisibility law of HCF and LCM:

Formula Used: HCF(a, b) must always divide LCM(a, b)

Checking each choice against 3600: $\frac{3600}{600} = 6$, $\frac{3600}{400} = 9$, $\frac{3600}{150} = 24$, whereas $\frac{3600}{500} = 7.2 \notin \mathbb{Z}$. Hence $\boxed{500}$ cannot be their HCF. [1M]

Q6. **Assertion (A):** If $\text{LCM}(a, b) = 182$ and $a \times b = 2366$, then $\text{HCF}(a, b) = 13$.

Reason (R): For any two positive integers a and b, $\text{HCF}(a, b) = \dfrac{a \times b}{\text{LCM}(a, b)}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

We know:

Formula Used: $\text{HCF}(a, b) = \dfrac{a \times b}{\text{LCM}(a, b)}$

Reason (R) is true by theorem. Computing $\frac{2366}{182} = \frac{182 \times 13}{182} = 13$ confirms Assertion (A). Hence both are true and (R) explains (A). [1M]

SECTION B: Very Short Answer Type

MATHS MANTHAN | **($2 \times 2 = 4$ Marks)**

Q7. Prove that $\dfrac{1}{\sqrt{2}}$ is irrational, given that $\sqrt{2}$ is irrational. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $\sqrt{2}$ is irrational. Suppose, on the contrary, that $\frac{1}{\sqrt{2}} = \frac{a}{b}$ $(a, b \in \mathbb{Z},\ b \neq 0$, with $a \neq 0$ since $\frac{1}{\sqrt{2}} \neq 0)$. Inverting both sides:

$$\sqrt{2} = \frac{b}{a}. \qquad [1M]$$

We know the reciprocal property of non-zero rationals:

Formula Used: $a, b \in \mathbb{Z} \setminus \{0\} \implies \dfrac{b}{a} \in \mathbb{Q}$

This implies $\sqrt{2}$ is rational, contradicting the given fact that $\sqrt{2}$ is irrational. Hence $\boxed{\dfrac{1}{\sqrt{2}} \text{ is irrational.}}$ [1M]

Q8. Prove that $7\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is irrational. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $\sqrt{5}$ is irrational. Suppose, on the contrary, that $7\sqrt{5} = \frac{a}{b}$ is rational $(a, b \in \mathbb{Z},\ b \neq 0)$. Dividing both sides by 7:

$$\sqrt{5} = \frac{a}{7b}. \qquad \text{[1M]}$$

We know rational numbers are closed under non-zero integer division:

Formula Used: $a, b \in \mathbb{Z},\ 7b \neq 0 \implies \frac{a}{7b} \in \mathbb{Q}$

Hence $\sqrt{5}$ is rational, contradicting that $\sqrt{5}$ is irrational. Therefore $\boxed{7\sqrt{5} \text{ is irrational.}}$ **[1M]**

SECTION C: Short Answer Type

MATHS MANTHAN | **(2 × 3 = 6 Marks)**

Q9. Prove that $\sqrt{p}$ is irrational for any prime p. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Assumption: Suppose, to the contrary, that $\sqrt{p}$ is rational for a prime p. Then there exist co-prime integers a, b $(b \neq 0, \text{HCF}(a, b) = 1)$ such that:

$$\sqrt{p} = \frac{a}{b} \implies a^2 = pb^2. \qquad \text{— (1)} \qquad \text{[1M]}$$

We know the Fundamental Theorem of Arithmetic divisibility theorem:

Formula Used: If a prime p divides a^2, then p divides a.

From (1), $p \mid a^2 \implies p \mid a \implies a = pc$ for some integer c. Substituting into (1):

$$(pc)^2 = pb^2 \implies p^2c^2 = pb^2 \implies b^2 = pc^2. \qquad \text{[1M]}$$

Thus $p \mid b^2 \implies p \mid b$. Now p divides both a and b, contradicting $\text{HCF}(a, b) = 1$. Hence $\boxed{\sqrt{p} \text{ is irrational for any prime } p.}$ **[1M]**

Q10. If the HCF of 210 and 55 is expressible in the form $210 \times 5 + 55y$, find the value of y. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

First finding HCF$(210, 55)$ via prime factorisation (or Euclid's division: $210 = 55 \times 3 + 45$, $55 = 45 \times 1 + 10$, $45 = 10 \times 4 + 5$, $10 = 5 \times 2 + 0$):

$$210 = 2 \times 3 \times 5 \times 7, \qquad 55 = 5 \times 11 \implies \text{HCF}(210, 55) = 5. \qquad \textbf{[1.5M]}$$

Equating the given expression to HCF$(210, 55) = 5$:

Formula Used: $210 \times 5 + 55y = \text{HCF}(210, 55)$

Solving step by step:

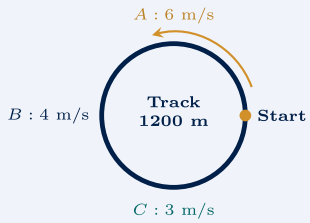
$$1050 + 55y = 5 \implies 55y = 5 - 1050 = -1045 \implies y = \frac{-1045}{55} = \boxed{-19}. \qquad \textbf{[1.5M]}$$

SECTION D: Long Answer Type

MATHS MANTHAN | **(2 × 5 = 10 Marks)**

Q11. Three athletes A, B, and C start running around a circular track of circumference 1200 m at speeds of 6 m/s, 4 m/s, and 3 m/s respectively from the same starting point and in the same direction. (i) Find the time taken by each athlete to complete one full lap. (ii) After how many seconds will all three athletes meet again at the starting point? (iii) How many laps will each athlete have completed by that time? **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Circular Reunion: Because all three run in the same direction from the common starting line along a 1200 m circuit, they reunite at the start after LCM$(200, 300, 400) = 1200$ s.

We know time-speed-distance and simultaneous return formulas:

Formula Used: $\text{Time} = \dfrac{\text{Distance}}{\text{Speed}}, \qquad T_{\text{meet}} = \text{LCM}(T_A, T_B, T_C)$

(i) Time for one lap:

$$T_A = \frac{1200}{6} = \boxed{200 \text{ s}}, \qquad T_B = \frac{1200}{4} = \boxed{300 \text{ s}}, \qquad T_C = \frac{1200}{3} = \boxed{400 \text{ s}}. \qquad \textbf{[2M]}$$

(ii) Time to meet together at starting point: Factorising $200 = 2^3 \times 5^2$, $300 = 2^2 \times 3 \times 5^2$, $400 = 2^4 \times 5^2$:

$$\text{LCM}(200, 300, 400) = 2^4 \times 3 \times 5^2 = 16 \times 75 = \boxed{1200 \text{ seconds (or 20 minutes)}}. \qquad \textbf{[1.5M]}$$

(iii) Laps completed by each athlete:

$$\text{Laps by } A = \frac{1200}{200} = \boxed{6}, \qquad \text{by } B = \frac{1200}{300} = \boxed{4}, \qquad \text{by } C = \frac{1200}{400} = \boxed{3}. \qquad \textbf{[1.5M]}$$

Q12. In an industrial plant, three automated sirens sound at intervals of 15 minutes, 20 minutes, and 25 minutes respectively. A fourth master siren sounds every 30 minutes. If all four sirens sound simultaneously at 6:00 AM: (i) Find the exact time when they will next sound simultaneously. (ii) How many times will all four sound simultaneously between 6:00 AM and 6:00 PM (including both endpoints)? (iii) Which siren sounds the greatest number of times during this 12-hour period, and how many times? [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

We know synchronization interval equals LCM of individual periods:

Formula Used: $15 = 3 \cdot 5,\ 20 = 2^2 \cdot 5,\ 25 = 5^2,\ 30 = 2 \cdot 3 \cdot 5 \implies \text{LCM} = 2^2 \times 3 \times 5^2 = 300$ min

(i) Next simultaneous time: 300 min $= \frac{300}{60} = 5$ hours. Adding 5 hours to 6:00 AM gives $\boxed{\text{11:00 AM}}$. [2M]

(ii) Simultaneous tolls from 6:00 AM to 6:00 PM: Over 12 hours = 720 min, multiples of 300 min inside $[0, 720]$ occur at 0 min (6:00 AM), 300 min (11:00 AM), and 600 min (4:00 PM). Total = $\boxed{\text{3 times}}$. [1.5M]

(iii) Most frequent siren: The shortest-interval siren ($\boxed{\text{Siren 1, interval 15 min}}$) sounds most often. Including 6:00 AM and 6:00 PM:

$$\text{Tolls of Siren 1} = \frac{720}{15} + 1 = 48 + 1 = \boxed{\text{49 times}}. \qquad [1.5\text{M}]$$

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Q13. CASE STUDY 10: OFF-GRID SOLAR ENERGY STORAGE BATTERY CHARGE CYCLES [4 Marks]

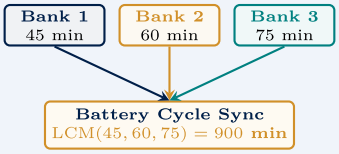
An off-grid solar-powered eco-resort in Ladakh uses three banks of lithium-iron-phosphate ($LiFePO_4$) battery arrays to store solar photovoltaic energy. Due to varying capacities and charging regulators, Bank 1 completes a full charge-discharge cycle in 45 minutes, Bank 2 in 60 minutes, and Bank 3 in 75 minutes. The plant supervisor synchronizes all three battery banks to start charging at 7:00 AM.

(i) Find the least common multiple (LCM) of $45, 60$, and 75 minutes. [2]

(ii) After how many hours from 7:00 AM will all three battery banks complete their cycles simultaneously again? [1]

(iii) Express 75 as a product of its prime factors in exponential form. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Energy Cycle Alignment: Bank 1 finishes 20 cycles, Bank 2 finishes 15 cycles, and Bank 3 finishes 12 cycles when 900 min (15 hours) elapses at 10:00 PM.

We know the simultaneous cycle completion period is the LCM of individual cycle times:

Formula Used: Synchronization Period = LCM$(45, 60, 75)$

(i) LCM of $45, 60$, and 75: Prime factorising:

$$45 = 3^2 \times 5, \qquad 60 = 2^2 \times 3 \times 5, \qquad 75 = 3 \times 5^2.$$

Taking highest powers of $2, 3, 5$:

$$\text{LCM}(45, 60, 75) = 2^2 \times 3^2 \times 5^2 = 4 \times 9 \times 25 = \boxed{900 \text{ minutes}}. \qquad [2M]$$

(ii) Elapsed hours from 7:00 AM: Converting 900 minutes into hours:

$$\frac{900}{60} = \boxed{15 \text{ hours}} \quad \text{(i.e., at 10:00 PM).} \qquad [1M]$$

(iii) Prime factorisation of 75: $75 = 3 \times 25 = \boxed{3 \times 5^2}$. [1M]

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